



Test Planning and Documentation Improvement Reflection

Assignment Context

This document addresses the requirements of Assignment 4, which asks for a critical review and enhancement of previous test planning and documentation work, using the best practices from Module 4 (Formal Test Planning, Documentation, Risk Analysis, and Ethical G-AI Usage).

1. Original Work Reviewed

- **Reviewed Artifact:** [Integration and System Testing Report \(Part 1\)](#)
- **Reason for Selection:** This report comprehensively documented integration and system testing for the MobileFoodDeliveryApp, including test strategies, scenarios, code, and lessons learned. It is a suitable foundation for applying new best practices.

2. Critical Self-Assessment

Strengths of the Original Submission

- Clear documentation of integration and system testing activities.
- Inclusion of both functional and non-functional tests.
- Evidence of test execution (code, logs, screenshots).
- Reflection on challenges and lessons learned.

Areas for Improvement (per Module 4)

- **Test Plan Structure:** Lacked a formal, standalone test plan section (objectives, scope, resources, schedule, risk analysis).
- **Risk Register:** No explicit risk register or contingency planning.
- **Documentation Format:** Test cases could be further standardized (e.g., YAML/Markdown tables).
- **Test Data:** Test data was present but not always clearly separated or documented.
- **Ethical G-AI Usage:** No mention of G-AI assistance or validation protocols.

3. Improvements Made

3.1. Formal Test Plan (Added Below)

- **Objectives:** Clearly defined for both integration and system testing.
- **Scope:** Explicitly stated what is in and out of scope.
- **Resources:** Listed team roles, tools, and environments.
- **Schedule:** Outlined test phases and milestones.
- **Risk Analysis:** Added a risk register and contingency plan.

3.2. Documentation Enhancements

- Standardized test case documentation using Markdown tables.
- Added a sample YAML test case for illustration.
- Clarified and separated test data where relevant.

3.3. G-AI Usage and Validation

- Used G-AI (GitHub Copilot) to brainstorm improvements and structure.
- All AI-generated suggestions were reviewed and customized to fit project context.

4. Formal Test Plan (MobileFoodDeliveryApp)

4.1. Objectives

- Validate integration between UI, order placement, and payment modules.
- Ensure system meets functional and non-functional requirements (performance, usability).

4.2. Scope

- **In Scope:**
 - Integration of `main.py`, `Order_Placement.py`, `Payment_Processing.py`
 - Functional and non-functional system tests (performance, usability)
- **Out of Scope:**
 - Security testing
 - Third-party service integration (beyond stubs)

4.3. Resources

- **Team:** Test Lead, Integration Specialist, System Tester
- **Tools:** Python, unittest, threading, manual usability checklist
- **Environment:** Local development machine (Linux), Python 3.10

4.4. Schedule

Phase	Start Date	End Date	Milestone
Test Planning	10/10/2025	12/10/2025	Test plan completed
Integration Testing	13/10/2025	15/10/2025	Integration tests passed
System Testing	16/10/2025	18/10/2025	System tests passed
Reporting & Review	19/10/2025	20/10/2025	Submission

4.5. Risk Register

```
risks:  
- id: R001  
  description: Delay in test environment setup  
  impact: High  
  likelihood: Medium  
  contingency: "Regular environment checks; backup environment ready; reschedule tests"  
- id: R002  
  description: Incomplete test data  
  impact: Medium  
  likelihood: Medium  
  contingency: "Review test data before execution; peer review of test cases."  
- id: R003  
  description: Unclear requirements for edge cases  
  impact: Medium  
  likelihood: High  
  contingency: "Consult with stakeholders; document assumptions; update tests as needed"
```



5. Sample Standardized Test Case (YAML)

```
test_case:
  id: TC1
  title: Place order with valid cart and payment
  objective: Ensure order is confirmed and payment processed
  preconditions:
    - User is logged in
    - Cart contains at least one valid item
  steps:
    - Navigate to checkout
    - Enter valid payment details
    - Submit order
  expected_result: Order confirmation and payment success message
```

6. Reflection

Assignments/Projects Revisited

- Integration and System Testing Report (Part 1) for MobileFoodDeliveryApp

Changes Made

- Added a formal test plan section with objectives, scope, resources, schedule, and risk register.
- Standardized test case documentation and clarified test data.
- Used G-AI to structure improvements, with all outputs reviewed for accuracy and context.

Challenges and Solutions

- **Challenge:** Original report lacked formal risk analysis and standardized documentation.
- **Solution:** Applied Module 4 best practices to introduce these elements, improving clarity and project readiness.

Value of New Methods

- The new structure ensures better planning, risk management, and documentation quality, making the testing process more robust and auditable.

7. Git Commit Reference

- Commit: 0970992

End of Reflection and Improvements